Gitesh Chawda

Machine Learning Engineer
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PROFESSIONAL SUMMARY

Engineer with experience spanning deep learning research, LLM pretraining, GPU optimization, and production software development. Pretrained a 247M-parameter language model on 25.5B tokens and developed multi-backend ML infrastructure supporting 118 model architectures across JAX, PyTorch, and TensorFlow. Experienced in GPU kernel benchmarking, numerical correctness evaluation, and building secure, scalable financial applications using Java, React, GraphQL, and AWS.

TECHNICAL SKILLS

Languages and Web: Python, C++, TypeScript, JavaScript, SQL, React.js, FastAPI, GraphQL, NumPy, Pandas, OpenCV
ML and Deep Learning: PyTorch, TensorFlow, Keras 3, JAX, Hugging Face Transformers, Computer Vision, Diffusion Models
LLM Training and Inference: vLLM, TensorRT-LLM, Triton, CUDA, XLA, Quantization (INT8/INT4/FP8), LoRA, PEFT, SFT, GRPO, KV Cache Optimization, FlashAttention, Model Serving
Agentic AI: LangChain, LangGraph, LlamaIndex, CrewAI, OpenAI Agents SDK, Retrieval-Augmented Generation (RAG), MCP
MLOps and Cloud: Docker, Linux, Git, GitHub Actions, CI/CD, AWS (Lambda, API Gateway, S3), GCP, MongoDB, Neo4j, Chroma

EXPERIENCE

AI Software Engineer, AfterQuery Experts (Contract) | Remote *Sep 2026 - Present*

- Benchmarked Muon optimizer kernels on NVIDIA H100 GPUs using correctness oracles and latency baselines, achieving 3.85–4.17× speedup and 74–76% lower latency while validating BF16 Newton–Schulz accuracy.
- Created PyTorch, CUDA, and Triton MoE dispatch/combine tests for top-k routing, expert capacity, token packing, and FP32 accumulation, with reference-output checks and calibrated speedup scoring.
- Stress-tested multi-query attention during decoding, identifying causal masking and ragged KV-cache memory-safety edge cases with compute-sanitizer checks.
- Authored and reviewed AI coding-agent tasks for Projects Corgi and Gold, assessing algorithmic efficiency, API compatibility, concurrency safety, and regression coverage; ranked among Project Corgi's top 40 contributors.

Software Developer, Fiserv | Chennai, India *Jul 2023 - Aug 2024*

- Delivered secure banking and payments applications using React.js, GraphQL, Java, Spring Boot, and Oracle SQL; integrated reusable frontend components with AWS AppSync, Lambda, API Gateway, Cognito, and S3.
- Enforced server-side validation, authorization checks, and client-side tampering protections across banking workflows, reducing unauthorized access issues by 85%.
- Remediated 25+ critical security vulnerabilities identified through AppSec audits and Burp Suite testing, strengthening API access controls and application security compliance.
- Established and supported disaster recovery mechanisms for production banking applications to improve service availability, operational reliability, and continuity during disruptions.
- Managed 20+ production deployments and hotfixes using Jenkins, JIRA, QTest, and Splunk; analyzed production logs and coordinated with QA and engineering teams to resolve release incidents.

OPEN SOURCE MAINTAINER AND CONTRIBUTOR

- Created and solely maintain **ZeroModels**, a Keras 3 ecosystem of 118 LLM, VLM, vision, speech, diffusion, and depth model families across JAX, PyTorch, and TensorFlow.
- Enabled Hugging Face-to-Keras weight conversion with parity validation and INT8/INT4/FP8 quantization; reached 140K+ downloads across 700+ model repositories.
- Hugging Face Transformers: enabled batched Qwen2.5/3-Omni audio generation, fixed Hunyuan-VL preprocessing parity, decoupled LXMERT and ConvBERT tokenizers, and added repository-wide type hints.
- Keras and KerasCV: contributed multi-backend operations, PSNR, augmentation layers, ConvMixer, CIoU loss, and YOLOv8 improvements.

PROJECTS

- **LaMoE:** Designed and pretrained a 247.65M-parameter LLM with MLA, Gated DeltaNet, and MoE on 25.5B tokens, followed by SFT; built tokenizer, training, inference, and evaluation pipelines.
- **txtcode:** Open-source developer tooling project focused on working with code through text-based workflows.
- **ZeroInfer:** Local-first inference platform supporting 143+ model families with an OpenAI-compatible API and MCP server.
- **OpenRAG:** Self-hosted RAG with Chroma vector and Neo4j graph retrieval, cross-encoder reranking, and local LLM inference.

EDUCATION

Master of Science in Computer Science, California State University, Fullerton, CA *Aug 2024 - May 2026*
Bachelor of Engineering in Computer Science, Pimpri Chinchwad College of Engineering, Pune, India *Aug 2019 - Jul 2023*

PUBLICATIONS

EffiConvRes: An Efficient CNN with Residual Connections and Depthwise Convolutions. IEEE ICCUBEA, 2023.
Survey and Comparison of Various Pre-trained CNN Architectures and CNN-Transformer Combinations. AIP Conference Proceedings, 2024.